Smith's method of analysing aggressive contests, and apply it to sex.* It will be a little bit more complicated than the case of the hawks and doves, because we shall have two female strategies and two male strategies.

As in Maynard Smith's studies, the word 'strategy' refers to a blind unconscious behaviour program. Our two female strategies will be called *coy* and *fast*, and the two male strategies will be called *faithful* and *philanderer*. The behavioural rules of the four types are as follows. Coy females will not copulate with a male until he has gone through a long and expensive courtship period lasting several weeks. Fast females will copulate immediately with anybody. Faithful males are prepared to go on courting for a long time, and after copulation they stay with the female and help her to rear the young. Philanderer males lose patience quickly if a female will not copulate with them straight away: they go off and look for another female; after copulation too they do not stay and act as good fathers, but go off in search of fresh females. As in the case of the hawks and doves, these are not the only possible strategies, but it is illuminating to study their fates nevertheless.

Like Maynard Smith, we shall use some arbitrary hypothetical values for the various costs and benefits. To be more general it can be done with algebraic symbols, but numbers are easier to understand. Suppose that the genetic pay-off gained by each parent when a child is reared successfully is +15 units. The cost of rearing one child, the cost of all its food, all the time spent looking after it, and all the risks taken on its behalf, is −20 units. The cost is expressed as negative, because it is 'paid out' by the parents. Also negative is the cost of wasting time in prolonged courtship. Let this cost be −3 units.

Imagine we have a population in which all the females are coy, and all the males are faithful. It is an ideal monogamous society. In each couple, the male and the female both get the same average pay-off. They get +15 for each child reared; they share the cost of rearing it (−20) equally between the two of them, an average of −10 each. They both pay the −3 point penalty for wasting time in prolonged courtship. The average pay-off for each is therefore + 15 − 10 − 3 = + 2.

Now suppose a single fast female enters the population. She does very well. She does not pay the cost of delay, because she does not indulge in prolonged courtship. Since all the males in the population are faithful, she can reckon on finding a good father for her children